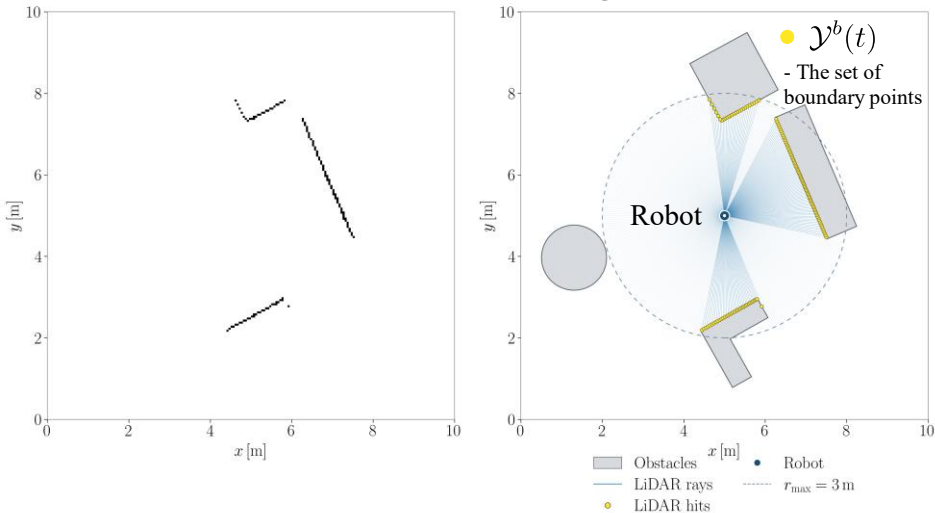


LiDAR sensing



$$\mathbf{y}^b \in \mathcal{Y}^b(t)$$

Online Error Bound Estimation

- Obstacle boundary measurements from onboard perception sensors (e.g. LiDAR)

$$e_h(\mathbf{y}^b, t) \approx \hat{e}_h(\mathbf{y}, t) + \hat{e}_{\nabla h}(\mathbf{y}, t)^\top (\mathbf{y}^b - \mathbf{y}),$$

$$\min_{\hat{e}_h, \hat{e}_{\nabla h}} \sum_{\mathbf{y}^b \in \mathcal{Y}^b(t)} \left| \underbrace{e_h(\mathbf{y}^b, t)}_{=-\hat{h}(\mathbf{y}^b, t)} - \hat{e}_h - \hat{e}_{\nabla h}^\top (\mathbf{y}^b - \mathbf{y}) \right|^2$$

$$\epsilon_h(\mathbf{y}, t) \approx |\hat{e}_h| + \delta_h, \quad \epsilon_{\nabla h}(\mathbf{y}, t) \approx \|\hat{e}_{\nabla h}\| + \delta_{\nabla h},$$